Do Students Agree with LLM Feedback on Their Own Code? A Within-Subject Study of Four Feedback Configurations in Software Project Courses

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Abstract. Oral defense (viva) in software engineering capstone projects is essential for verifying students' comprehension of business logic and codebase architecture, yet manual grading is labor-intensive and inconsistent, whereas off-the-shelf large language models (LLMs) risk code-context hallucinations, prompt injection, and sycophancy. This paper presents the pedagogical evaluation engine of **PACE**, an AI-assisted viva system integrating complexity-aware hybrid ground-truth extraction (routing raw code versus Mermaid control/data-flow graphs at node threshold $\tau = 10$) with three-layer guardrails and XML-based delimiter sandboxing. To mitigate single-model sycophancy, we propose a role-decoupled **Multi-LLM Ensemble** (*Researching* mode) chaining **DeepSeek-V3** (*Code Inspector*) with **Groq (`openai/gpt-oss-120b`)** (*Pedagogical Synthesizer and Rubric Grader*), evaluated alongside three single-LLM formative paradigms (*Sandwich*, *Rubric*, and *Explainable*). In a randomized, blinded, within-subject experiment at FPT University, 86 students across 28 repositories submitted 344 scenario-based viva responses rated on a nine-dimension, five-point Likert instrument adapted from the Feedback Perceptions Questionnaire (FPQ) and formative assessment frameworks. Across 344 instances, ratings endorsed AI judgment (79.9%), future use (68.0%), and motivation (41.9%), with a multidomain composite mean of 4.07/5 (95% CI [3.90, 4.23]). All six 95% repository-bootstrap contrasts on direct Acceptance included zero without establishing equivalence, and observational raw-versus-graph Acceptance remained inconclusive ($b = 0.03$, 95% CI $[-0.18, 0.26]$). Exploratory *Sandwich–Explainable* actionability differed by 0.37 points (simultaneous 95% sensitivity band [0.02, 0.72] across 90 contrasts) without isolating next-step effects. These findings concern reported perceptions rather than expert-verified correctness, observed revision, or learning gains.

Keywords: Automated Oral Assessment · Multi-LLM Ensemble · Formative Feedback · Pedagogical AI Guardrails · Software Engineering Education.

1 Introduction

This document serves as the standardized LaTeX template prepared according to Springer Computer Science Proceedings guidelines. Section headings are capitalized in Title Case (e.g., nouns, verbs, and major words capitalized).

Introduce the broader context, motivation, and practical challenges here. For instance, explaining the role of generative AI and automated assessment tools in software engineering education.

1.1 Research Questions

State the guiding research questions of your study clearly:

- **RQ1:** How do students perceive the quality and acceptance of AI-generated feedback (accuracy, clarity, specificity, and actionability)?
- **RQ2:** How does student agreement differ across distinct types or styles of AI feedback?
- **RQ3:** How does student perception vary between question generation sources (e.g., codebase-derived vs. knowledge-graph-derived)?

1.2 Contributions and Paper Structure

Summarize the key contributions of the paper:

1. **Contribution 1:** Outline the first major conceptual or technical contribution.
2. **Contribution 2:** Outline the empirical dataset and evaluation methodology.
3. **Contribution 3:** Discuss the practical implications for intelligent tutoring systems.

The remainder of this paper is structured as follows: Section 2 reviews related literature; Section 3 presents the methodology and architecture; Section 4 details the experimental evaluation; Section 5 discusses findings; Section 6 examines threats to validity; and Section 7 concludes the paper.

2 Related Work

Discuss relevant prior work in the literature. Citations are specified using square brackets and consecutive numbers. For single citations, use [4]. Multiple citations should be grouped and ordered numerically, such as [1,2,3]. When author names are used in the text, write: Foster and Kesselman [2] demonstrated that...

2.1 AI-Assisted Education and Feedback

Review existing approaches in automated feedback generation, large language models (LLMs), and cognitive load reduction in programming education.

2.2 Evaluation of Feedback Quality

Discuss prior frameworks evaluating constructive feedback, accuracy, transparency, and student receptivity.

3 Proposed Methodology and System Architecture

Detail the proposed framework, system design, data pipelines, or algorithmic foundations.

3.1 Architectural Overview

Equations must be numbered consecutively within parentheses at the right margin without section prefixes:

$$S(x, y) = \alpha \cdot A(x) + (1 - \alpha) \cdot B(y) \tag{1}$$

where $S(x, y)$ represents the composite score, and $\alpha \in [0, 1]$ is a balancing parameter.

3.2 Visual Representation and Accessibility

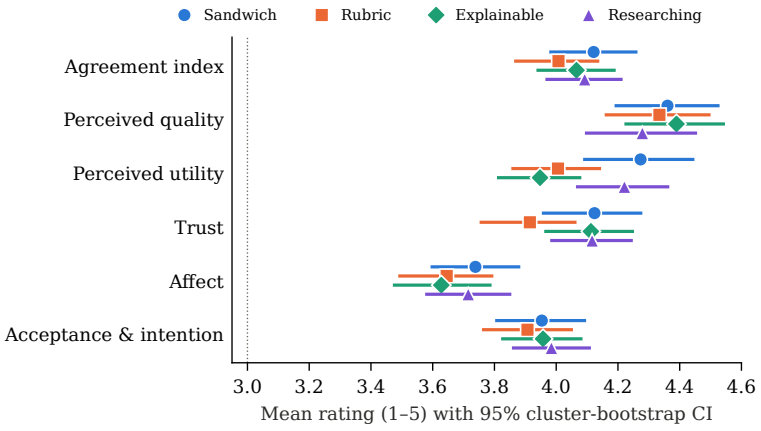


Fig. 1. Overview of the conceptual construct framework and evaluation metrics.

As shown in Fig. 1, all illustrations must be referenced directly in the text. Captions are set in 9-point font, centered if single-line, and justified if multi-line.

4 Empirical Evaluation and Survey Design

Describe the experimental setting, participant demographics, questionnaire instrumentation, and procedure.

4.1 Instrumentation and Construct Measurements

Table 1 highlights the evaluation dimensions. The numbering of tables and figures continues sequentially throughout the manuscript.

Table 1. Summary of survey constructs, dimensions, and operational definitions.

Construct	Metric/Code	Operational Definition
Accuracy	ACC	Correctness of technical explanations and error diagnoses.
Clarity	CLA	Ease of comprehension, structure, and readability.
Specificity	SPE	Concrete pointers to lines of code or specific logic.
Transparency	TRA	Explainability of why and how the feedback was derived.
Actionability	ACT	Practical guidance on how to fix or refine the implementation.
Acceptance	ACP	Student agreement and willingness to adopt recommendations.

5 Results and Discussion

Present statistical analyses, survey findings, correlation matrices, and thematic student responses.

5.1 Analysis of Survey Constructs (RQ1)

Present findings addressing RQ1. Reference additional figures where necessary:

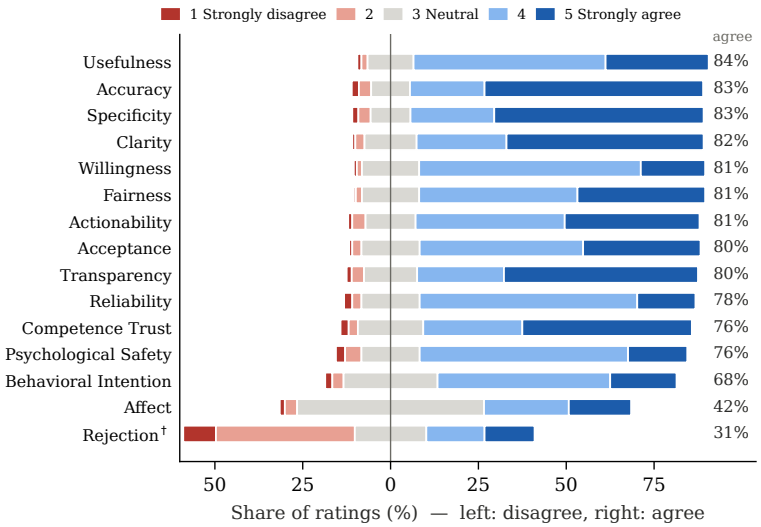


Fig. 2. Distribution of student responses across the Likert-scale items.

Fig. 2 illustrates the Likert-scale distributions observed in the participant cohort.

5.2 Cross-Feedback Comparative Analysis (RQ2)

Examine differences among feedback styles or presentation formats.

5.3 Source Modality Comparison (RQ3)

Compare question generation modalities (e.g., codebase extraction vs. knowledge graph).

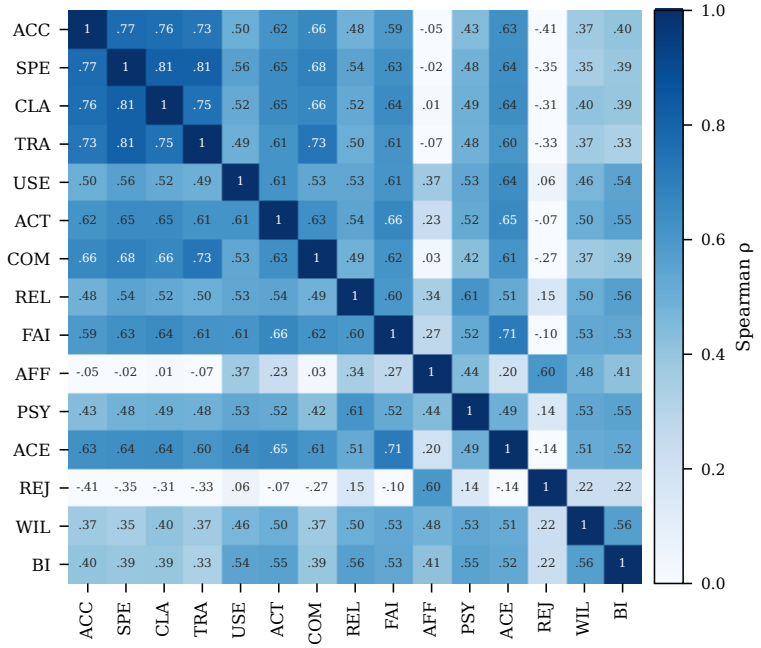


Fig. 3. Correlation matrix across evaluation dimensions and survey questions.

Fig. 3 depicts the inter-item correlation coefficients.

6 Threats to Validity

Discuss construct validity, internal validity, external validity, and reliability considerations.

7 Conclusion and Future Work

Summarize the main takeaways, lessons learned, and future research directions.

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Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

A Survey Instrument Details

Detailed survey questions and Likert-scale coding schemes can be provided here. Continues the same formatting standards as the main text.

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